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EUV ANALYSIS OF POLAR PLUMES

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(Received 5 January, 1977)

Abstract. Three polar plumes were studied using Skylab Mg x and O vi data. The plumes lie within the boundaries of a polar coronal hole. We find that the mean temperature of the plumes is about 1.1×10^6 K and that they have a small vertical temperature gradient. Densities are determined and found consistent with white light analyses. The variation of density with height in the plumes is compared with that expected for hydrostatic equilibrium. As is the case for other coronal features, polar plumes will be a source of solar wind if the magnetic field lines are open. On the basis of the derived plume model and estimates of the numbers of plumes in polar coronal holes, it appears that polar plumes contain about 15% of the mass in a typical polar hole and occupy about 10% of the volume.

1. Introduction

Polar plumes, which are believed to trace open magnetic field lines at the solar poles, have been intensively analyzed using white light eclipse photographs (cf. Van de Hulst, 1950; Nikolsky, 1956; Dziubenko, 1957; Saito, 1958, 1965; Newkirk and Harvey, 1968). These features have also been observed in extreme ultraviolet (EUV) emission lines by the Naval Research Laboratories (NRL) and Harvard experiments on Skylab (cf. Bohlin *et al.*, 1975a, b; Ahmad, 1976). Bohlin *et al.* (1975a, b) concluded that EUV polar plumes are the EUV counterparts of white light plumes because of their similar temperatures, density gradients, sizes and orientations. The results of Bohlin *et al.*, are based on the analysis of photographic observations acquired by the NRL Skylab experiment. Temperatures were inferred from the presence or absence of plumes in spectroheliograms of emission lines in the spectral region 250–610 Å. Density gradients were derived from Mg ix λ368 intensities.

The present paper reports results of an analysis of Mg x λ625 and O vi λ1032 observations of polar plumes obtained by the Harvard Skylab experiment. This experiment, which employed a photoelectric detection system, provided photometrically calibrated measurements with a spatial resolution of 5 arc sec $\times$ 5 arc sec (cf. Reeves *et al.*, 1976). These data have been utilized to derive temperature-density models for three polar plumes.

Determination of the physical conditions in polar plumes is important because these features are a major contributor to the XUV, white light and radio emission from polar coronal holes. Knowledge of the physical conditions in polar coronal

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Solar Physics **53** (1977) 397–408. *All Rights Reserved*
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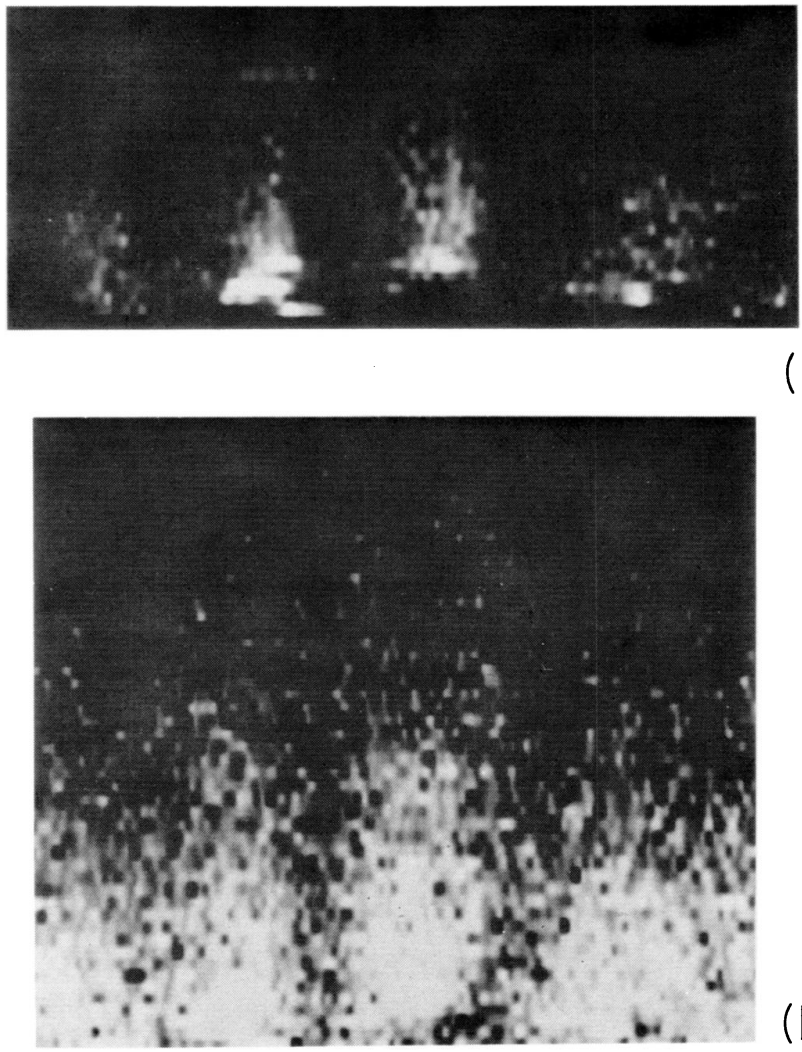
holes and features such as polar plumes, which are contained in these areas, is critical to the development of theories accounting for the association of high speed solar wind streams with coronal holes (cf. Krieger *et al.*, 1974; Neupert and Pizzo, 1974; Bell and Noci, 1976; Nolte *et al.*, 1976; Sheeley *et al.*, 1976). Because polar plumes appear to be associated with open field lines, they may contribute a significant amount of material to the solar wind originating from polar coronal holes.

2. Observations

As indicated above, polar plumes are observed at the solar poles in regions of low coronal density which have become known as coronal holes. Figure 1 shows a Mg x λ625 spectroheliogram of a 5 arc minute $\times$ 2 arc minute centered at the north pole. The pictures, which were made with different exposure times, were



POLAR PLUMES OBSERVED IN Mg x λ625

Fig. 1. Gray scale images of Mg x λ625 observations of northern polar plumes. NP1 is at center and NP2 to its left. (a) is a short exposure print slowing the distinctness of plumes. (b) is a longer exposure print on which the lower density outer regions of the plumes are visible.

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constructed from digital data using the Harvard Solar Image Display System. We have designated as NP1 the polar plume in the center of the picture and the plume on the left as NP2. The latter plume appears to lie over an EUV bright point located at a latitude of about 83° on the hemisphere of the sun facing the Earth. It appears that most, perhaps all, EUV polar plumes are rooted in EUV bright points (cf. Bohlin *et al.*, 1975a, b). EUV bright points, which are also prominent in X-ray photographs, are associated with small bipolar magnetic regions (cf. Golub *et al.*, 1974, 1976).

Three particularly well observed plumes were selected for analysis the plumes NP1 and NP2 mentioned above and another plume, designated NP1*, observed in the same location as NP1 eight hours later. NP1* may or may not be the same feature as NP1. Spectroheliograms were obtained simultaneously in the spectral lines Lyα λ1216, C II λ1335, C III λ977, O IV λ554, O VI λ1032, and Mg X λ625. The plumes were visible only in the latter two lines, which were used in the analysis described below. Data from 10 O VI an 10 Mg X spectroheliograms were utilized for the study of NP1 and NP2 while 6 spectroheliograms in each of these lines were utilized for NP1*. The individual spectroheliograms were summed in various combinations in order to avoid complications due to time evolution and to superimposed emission or absorption caused by macrospicules (cf. Withbroe *et al.*, 1976). Between the plumes can be seen emission due to various possible sources which we lump together and call the 'background'. The background consists of radiation from the quiet sun fore and aft of the polar coronal hole, from fainter plumes in the hole and from interplume material. This background was determined for the interstices between plumes and subtracted from the plume emission.

3. Temperature Analysis

The intensity of an EUV emission line is given by the relation (cf. Withbroe, 1970)

$$I = 1.73 \times 10^{-16} A f \int_{-\infty}^{\infty} g G(T) N_e^2 \, \mathrm{d}l, \tag{1}$$

where I is measured in erg cm^{-2} s^{-1} sterad^{-1}, A is the chemical abundance relative to hydrogen of the element forming the line, f is the oscillator strength, g is the Gaunt factor, $G(T)$ is a temperature dependent term depending on the excitation and ionization properties of the atom producing the line and l is the path length along the line of sight. Equation (1) assumes that the line emission results from collisional excitation by electrons. For O VI λ1032 there is also a contribution by resonance scattering of radiation from the chromospheric-coronal transition region. However, since the effects of resonance scattering do not affect our results substantially, we defer discussion of these effects until Section 5.

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Equation (1) also assumes ionization equilibrium. Effects arising from plasma outflow will also be discussed in Section 5.

If we assume that the gas along the line of sight is isothermal, the ratio of the intensities of the Mg x $\lambda 625$ and O vi $\lambda 1032$ lines will be (cf. Equation (1)).

$$\frac{I_{\mathrm{Mg\,x}}}{I_{\mathrm{O\,vi}}} = \frac{(AfgG(T))_{\lambda 625}}{(AfgG(T))_{\lambda 1032}}. \tag{2}$$

Because of differences between the temperature dependence of the functions $gG(T)$ for Mg x $\lambda 625$ and O vi $\lambda 1032$, the ratio of these lines is temperature sensitive (cf. Mariska and Withbroe, 1975). Hence, a comparison of theoretical and observed intensity ratios provides a measurement of the temperature of the plume under the assumption that the plume is isothermal. That assumption is consistent with the observed behavior of the Mg x/O vi ratio illustrated in Figure 2, which appears to remain constant across the plume within the errors of measurement. In evaluating the theoretical ratios we assumed $\log A_{\mathrm{O}} = 8.6$ and $\log A_{\mathrm{Mg}} = 7.5$ (on a scale where $\log A_{\mathrm{H}} = 12.0$), the oscillator strengths given by Wiese *et al.* (1966, 1969), and functions $gG(T)$ based on the Gaunt factors of Bely (1966) and ionization equilibrium calculations of Allen and Dupree (1969) and Dupree and Wood (1969).

The observed ratios $I_{\mathrm{Mg\,x}}/I_{\mathrm{O\,vi}}$ in the plumes vary from about 0.3 at 30 arc sec above the limb to about 0.8 between 100 and 125 arc sec above the limb, significantly below the value of 1.0 obtained for the quiet Sun at similar heights by

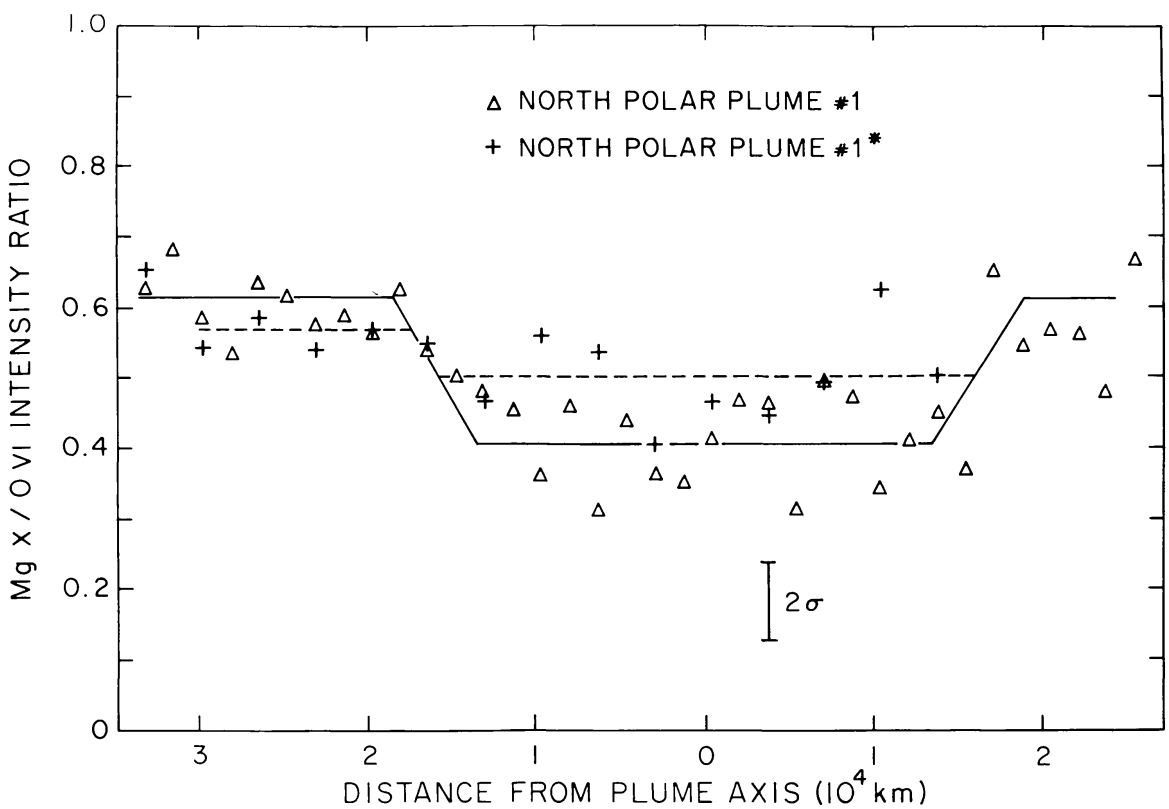

Fig. 2. Ratio of line intensities across the plume using intensities integrated between 1.06 and 1.09 $R_{\odot}$ above the limb. Average values inside and outside the plumes are indicated.

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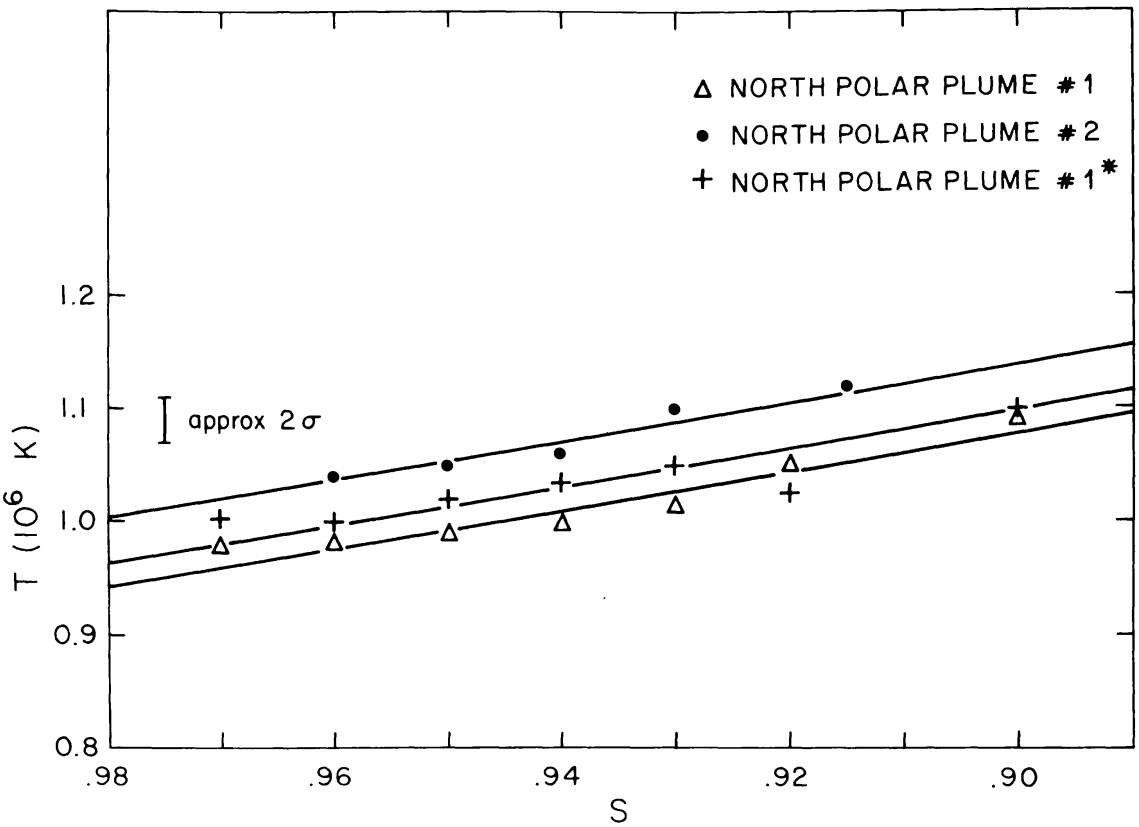

Fig. 3. Observed temperatures inferred from Mg x λ625/O VI λ1032 ratio and Allen–Dupree–Wood ionization equilibria. Solid lines are fits of lines of slope $\langle dT/dS \rangle = 1.6 \times 10^6$ to each plume.

Mariska and Withbroe (1975). This means that plumes must contain cooler material than the corona in the quiet Sun.

Figure 3 gives plume temperatures as a function of inverse radial distance, $S = R_\odot/R$ where $R_\odot$ is the radius of the Sun and R is the distance from Sun center to a point in the plume. As a first approximation the plume temperatures appear to increase linearly with decreasing S. The mean slope derived from measurements of the three plumes is $dT/dS = -1.6 \times 10^6$ K. The mean temperature of all three plumes for the range of heights observed is $T = 1.1 \times 10^6$ K. This compares favorably with the value of 1.2×10^6 K derived by Saito (1965) under the assumption that white light plumes are isothermal and in hydrostatic equilibrium. It also falls within the range of temperatures, 0.7 to 1.2×10^6 K, inferred by Bohlin *et al.* (1975a, b) from the presence or absence of polar plumes in spectroheliograms of various lines found between 250 and 610 Å.

4. Density Analysis

For simplicity we assumed a cylindrical model in which the electron density decreased exponentially with distance r from the axis of the cylinder,

$$N_e(R, r) = N_0(R) \exp\left(-\left(\frac{r}{\beta}\right)^2\right), \tag{3}$$

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where $N_0(R)$ is the density at the axis at a distance R from the center of the Sun. The parameter β provides a measurement of the effective width of the plume. An expression analogous to Equation (3) was used by Saito (1958) to study the variation of surface brightness across plumes observed in white light. An alternate relationship between the electron density and r is the one used by Newkirk and Harvey (1968),

$$N_e(r) = N_0\left(\frac{(d-r)}{d}\right)^{1.6}, \tag{4}$$

where d is the radius of the plume. Intensities computed with this relationship were found to fit the EUV data nearly as well as the exponential relationship. From the Earth one measures the EUV intensity as a function of distance ρ from the central axis of the plume. From Equations (1) and (3) and the geometrical relationship between ρ and r we obtain

$$I(R, \rho) = 3.46 \times 10^{-16} Af < gG(T) > N_0^2(R) \int_\rho^\infty \frac{\exp\left(-2\left(\frac{r}{\beta}\right)^2\right) dr}{\sqrt{r^2-\rho^2}}. \tag{5}$$

This equation assumes that the temperature in the plume does not vary across the width of the plume, an assumption consistent with the observed behavior of the Mg X/O VI ratio (see Section 3).

Figure 4 gives the parameter β as a function of distance above the limb. These widths were determined from the Mg X measurements. The widths obtained from the O VI data are identical to within the errors of the determinations. The uncertainty in the widths is about 2500 km. A least squares straight line has been fit to the data for the two plumes, NP1 and NP2. Both lines have the same slope and indicate that the plume widths increase with height. Since the widths measured for NP1* have such large scatter, we did not fit a least squares line to them. The overall trend suggests that the width of this plume also increases with height if the discrepant point at $1.07R_\odot$ is ignored. The widths for NP1* are less reliable than those for NP1 and NP2 due to the low contrast between the intensity of this plume and the background which makes it difficult to reliably determine the variation of intensity across the plume.

The increase in width as a function of height for these EUV plumes is larger than expected on the basis of an extrapolation of Saito's (1965) results for white light plumes observed in the 1962 New Guinea eclipse. Linear extrapolation of his results for $1.12R_\odot \leq R \leq 1.4R_\odot$ to lower heights leads one to expect that β would increase only slightly between $R = 1.0R_\odot$ and $R = 1.1R_\odot$, from about 2.7×10^4 km to 2.8×10^4 km. Because of the small number of EUV plumes studied, and because of the uncertainty of the legitimacy of extrapolating the white light results, we are unable to conclude if this difference is significant. The widths of our plumes are in good agreement with those of XUV plumes measured by Bohlin

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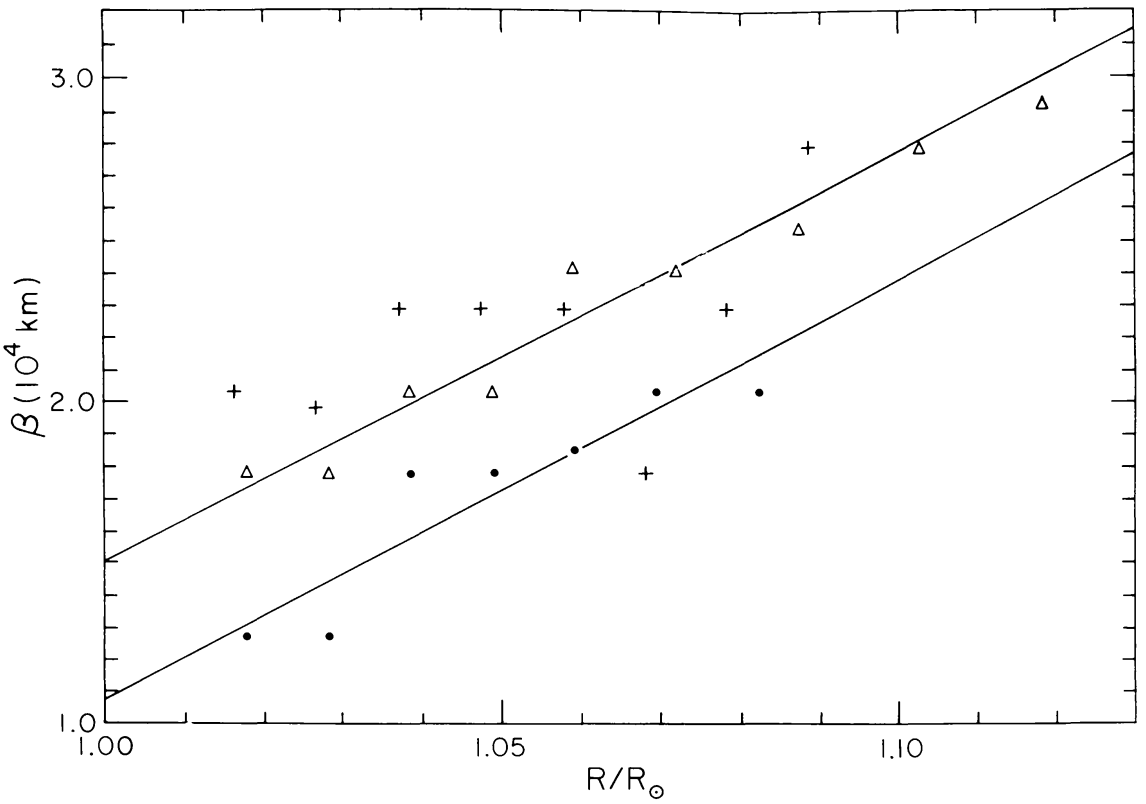

Fig. 4. Characteristic width of the density distributions as a function of height. Solid lines are straight line fits to NP1 and NP2. The triangles are for plume NP1, circles for NP2 and crosses for NP1*.

et al. (1975a, b) who obtained a mean width corresponding to $\beta = 3.2 \pm 0.8 \times 10^4$ km at a radial distance of $1.1 R_\odot$. Newkirk and Harvey (1968) obtained a similar result for white light plumes observed in eclipses of 1962, 1963, and 1965, a mean width corresponding to $\beta = 3.1 \pm 1.0 \times 10^4$ km for $R = 1.14 R_\odot$.

Figure 5 illustrates for three heights the fit of the theoretical Mg x intensities (curves) calculated from Equation (5) and the parameter β obtained from the mean relationship between β and R as given in Figure 4. For each theoretical curve an appropriate amount of background intensity has been added to the intensity contributed by the plume.

Through use of the widths β determined from the straight line fit to the data in Figure 4, the mean variation of temperature with height obtained from Figure 3, the measured central intensities of the plumes and Equation (5), the central densities $N_0(R)$ were determined as a function of R for each of the three plumes. The results are plotted in Figure 6. For comparison we have also plotted (dashed line) Saito's (1965) densities shifted downward by 0.08 dex to obtain a better fit to our results. His values, which were derived from white light plumes observed in 1962, are in good agreement with ours, although there is a small difference in density gradient and in absolute scale.

The solid line plotted in Figure 6 gives the variation of density with height at the plume axis predicted under the assumption that the plume is in hydrostatic

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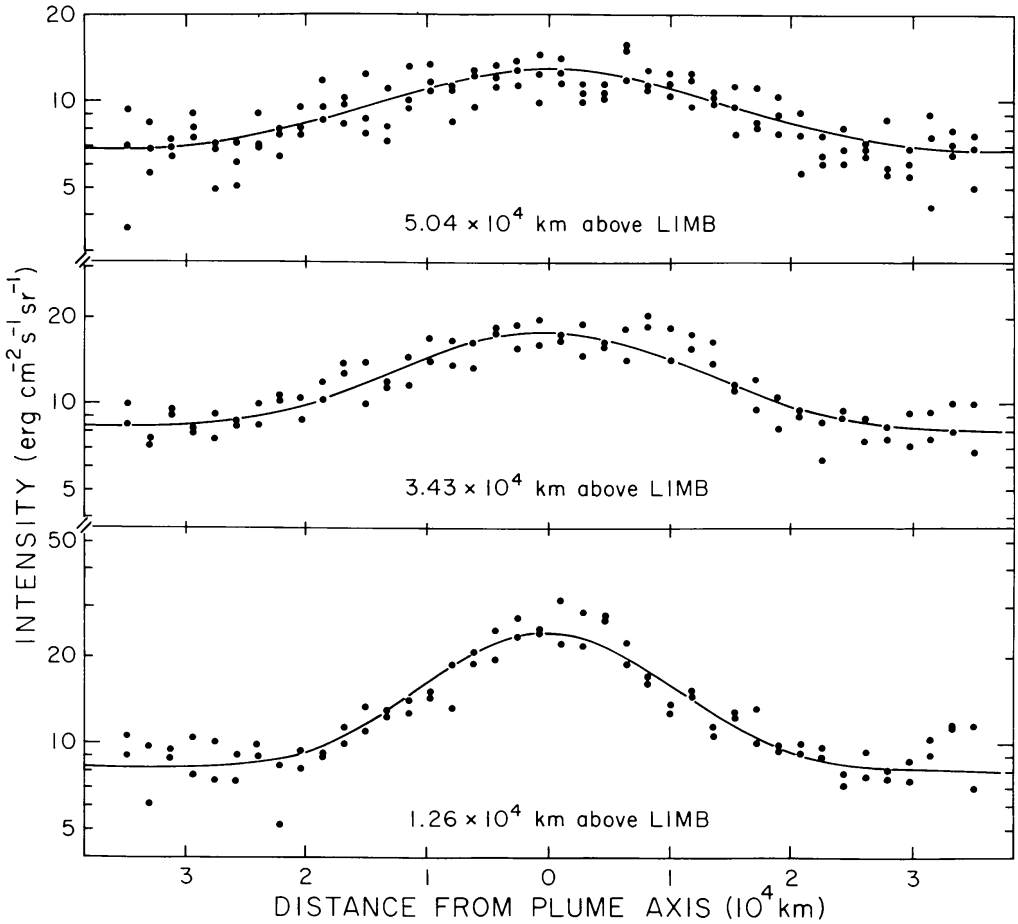

Fig. 5. Theoretical intensity distribution across a plume in Mg x for a plume with a Gaussian density distribution fit to the observations of NP1 at three heights.

equilibrium. One can show that if $\mathrm{d}T/\mathrm{d}S$ is constant, then the equation of hydrostatic equilibrium yields the relation,

$$\ln N_0(S)/N_0(S_0) = \left(1 - \frac{GM_\odot \mu M_\mathrm{H}}{kR_\odot \langle \mathrm{d}T/\mathrm{d}S \rangle}\right) \ln (T_\odot/T), \tag{6}$$

where N_0 is the density at the axis of the plume, μ is the mean molecular weight (assumed to be 0.62), T_0 is the temperature at reference height S_0 and the other parameters have then usual definitions. The solid line in Figure 6 was calculated from this equation and the derived relation between S and T (see Figure 3). Within the errors the data are consistent with the assumption of hydrostatic equilibrium over most of the observed height range. There is a slight departure of the densities for NP1 and NP2 from the hydrostatic equilibrium line, the significance of which will be discussed in the next paragraph.

If the plume is to be in hydrostatic equilibrium away from the plume axis, then the plume must be threaded by magnetic field lines that diverge from the plume axis with increasing height in the plume. Along each field line the ratio of the densities at two heights in the plume must be the same as the corresponding ratio along the plume axis. This implies (cf. Equation (3)) that the field lines at a height

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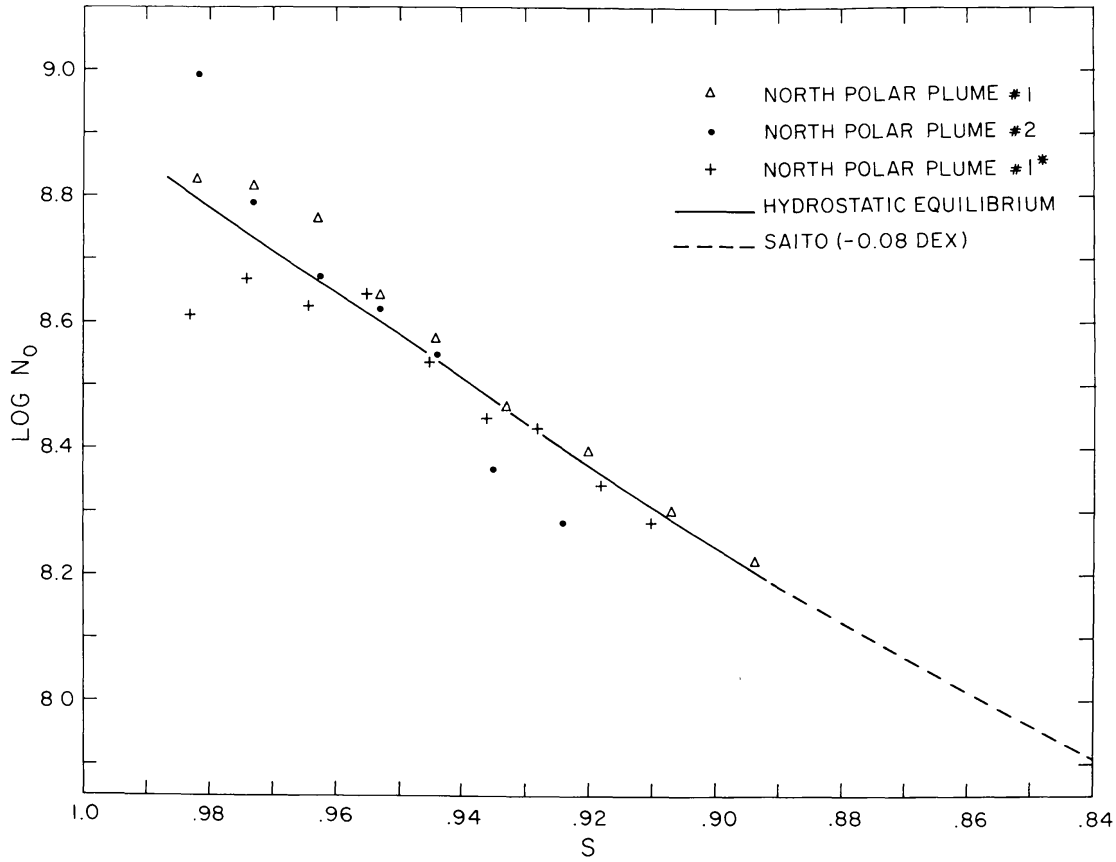

Fig. 6. Vertical density distribution at plume center.

corresponding to $S_2 = R_\odot/R_2$ will be a factor of $\beta(S_2)/\beta(S_1)$ further away from the plume axis than they are at a height corresponding to $S_1 = R_\odot/R_1$.

We wish to investigate the significance of the apparent departure of NP1 and NP2 from hydrostatic equilibrium. We shall make a hydrodynamic analysis of NP1, since it is the better observed of the two plumes. As in Saito's (1965) hydrodynamic analysis of the white light plumes, we employ the momentum equation,

$$v\frac{\mathrm{d}v}{\mathrm{d}R} = \frac{kT}{\langle m\rangle}\left(\frac{\mathrm{d}\ln N}{\mathrm{d}R} + \frac{\mathrm{d}\ln T}{\mathrm{d}R}\right) - \frac{GM_\odot}{R^2}, \tag{7}$$

where $\langle m\rangle$ is the mean particle mass, which we estimate to be 1.02×10^{-24} gms, appropriate to a fully ionized plasma, 90% H and 10% He by number. We write the continuity equation in the form,

$$\frac{\mathrm{d}\ln v}{\mathrm{d}R} = \frac{\mathrm{d}\ln(1/N_0\beta^2)}{\mathrm{d}R} \equiv X(R), \tag{8}$$

where X may be determined from the information in Figures 5 and 6. Substituting Equation (8) into Equation (7) we have

$$v^2 = -\frac{kT}{\langle m\rangle X}\left(\frac{\mathrm{d}\ln N}{\mathrm{d}R} + \frac{\mathrm{d}\ln T}{\mathrm{d}R}\right) - \frac{GM_\odot}{XR^2}. \tag{9}$$

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Assuming NP1 is in the plane of the sky we have at $R = 1.06 R_\odot$, $T = 1.03 \pm 0.03 \times 10^6$ K, $d \ln T/dR = 2.1 \pm 0.3 \times 10^{-11}$ cm^{-1}, $d \ln N/dR = 2.12 \pm 0.1 \times 10^{-10}$ cm^{-1}, and $X = 1.28 \pm 0.16 \times 10^{-10}$ cm^{-1}. Then the most probable value for v is 41 km s^{-1}. Using the 1σ extrema for the parameters quoted above, we get lower and upper limits for v of 0 km^{-1} s and 65 km s^{-1} respectively. A qualitatively similar case may be made for NP2. NP1* seems most consistent with hydrostatic equilibrium, but it has the worst scatter and carries the least weight. We conclude that while the observations are consistent with hydrostatic equilibrium within the errors of the analysis, they are in better agreement with a picture in which high velocity outflow is taking place. The effects of rapid outflow on the assumption of ionization equilibrium is considered in Section 5. From the integral of Equation (8) we can calculate the relative velocity through the region observed. We find that the velocity at $R = 1.12 R_\odot$ is about 35% greater than that at $1.06 R_\odot$ while the velocity at $R = 1.02 R_\odot$ is about 15% lower.

On the basis of the plume models derived in this investigation, it appears that plumes can contain a significant fraction of the total coronal material in a polar coronal hole. The density along the axis of a plume appears to be about a factor of about 3 higher than in typical polar hole models (Mariska, 1976; Munro, 1976). If one assumes our plume model is representative of an average plume and that the number of plumes in a polar hole is equal to the number of coronal bright points found there (cf. Golub *et al.*, 1974, 1976), plumes will cover about 10% of the area in a polar coronal hole and contain about 15% of the mass. We assume here that all coronal bright points in polar holes are associated with polar plumes, a hypothesis that appears to be consistent with the observations, as was mentioned earlier.

5. Effects of Resonance Scattering and Departures from Ionization Equilibrium

Noyes (1976) has found that resonance scattering of O VI λ1032 radiation from the transition layer can contribute a significant fraction of the radiation emitted by coronal material. We evaluated the effect of omitting the influence of resonance scattering on the analysis of plume observations and found that the plume temperatures, temperature gradients, densities, and widths (β) change by about 10% or less when resonance scattering is taken into account. If the plume plasma is in motion, the effects are even smaller because the coronal line profile is Doppler shifted with respect to the wavelength of the transition region line radiation. Since the magnitude of the flow velocities are uncertain and because the parameters derived for our plume model are uncertain by approximately 10% due to uncertainties in instrumental calibration, ionization equilibrium calculations, and the simplified plume model, etc., introduction of the effects of resonance scattering would not substantially improve the reliability of the plume model. Therefore, in the interest of simplicity, we have omitted the effects of resonance scattering in deriving the models.

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If material is flowing out at rapid velocities, departures from ionization equilibrium may be significant (see, e.g., Tworkowski, 1975). The character of such departures is not well enough understood at present to make quantitative predictions, but since the velocities would be increasing with height (see Section 4), we may say qualitatively that the effect is that the kinetic temperature will exceed the ionization temperature by an amount increasing with height in the low corona. In that case the absolute temperatures and the temperature gradient will be larger than derived in the present work.

6. Conclusions

Plumes are an important source of coronal emission in polar coronal holes. The plumes studied here have a vertical temperature gradient $dT/dS \cong -1.6 \times 10^6$ K, and an absolute temperature of about 1.1×10^6 K. The density distribution across the plume appears Gaussian, but it is also consistent with near-Gaussian distributions previously used to intrepret some white light observations. The density at plume center falls off a logarithmic gradient slightly steeper than that observed at higher altitudes in white light. The possibility that the plasma is in hydrostatic equilibrium cannot be ruled out, but flow velocities of the order of several times $10\ \mathrm{km\ s^{-1}}$ are in as good, if not better, agreement with the data. Various parameters we have obtained may be modified up to 10% by a correction for the effects of resonant scattering of the O VI line, but our general conclusions stand. It appears that polar plumes may contain about 15% of the mass in a typical polar coronal hole and occupy 10% of the volume.

Acknowledgements

We are grateful to J. D. Bohlin for some stimulating conversations. We thank H. Wadzinski for his help with the computer programming and M. Mouris for his assistance with the data analysis. We are particularly grateful to J. Mariska for his part in calculating the effects of resonant scattering. In carrying out this research, the authors have benefitted considerably from their participation in the Skylab Solar Workshop series on Coronal Holes. The Workshops are sponsored by NASA and NSF and are managed by the High Altitude Observatory. This work was supported in part by NASA contract NAS 5–3949 and Air Force Geophysical Laboratories contract F19628–76–C–0281.

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